

Commerce XI Maths App

Miscellaneous Exercise 4

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 4
Bivariate Frequency Distribution and Chi Square Statistic
Total questions: 24

Question 1 - MCQ · PG-53 · MEX-Q1

Following data gives the coded price (x) and demand (y) of a commodity.

Price	5	7	9	8	10	7	9	8	5	11	11	10	2	3	9
Demand	9	15	13	15	14	10	11	14	10	14	6	14	15	11	12
Price	2	4	3	14	6	10	7	15	8	6	5	6	11	14	15
Demand	6	11	8	11	10	15	9	15	13	9	14	10	7	5	6

Classify the data by taking classes 0–4, 5–9 etc. for x and 5–8, 9–12 etc. for y.

A.

$\begin{matrix} \text{X} \\ \text{Y} \end{matrix}$	0–4	5–9	10–14	15–19	Total
5–8	2	0	3	1	6
9–12	2	9	1	0	12
13–16	1	6	4	1	12
Total	5	15	8	2	30

B.

$\begin{matrix} \text{X} \\ \text{Y} \end{matrix}$	16–18	18–20	20–22	22–24	Total
10–20	2	0	0	0	2
20–30	0	0	0	0	0
30–40	0	0	0	5	5
40–50	3	2	3	1	9
50–60	2	1	5	1	9
60–70	0	2	1	2	5
Total	7	5	9	9	30

C.

$\begin{matrix} \text{X} \\ \text{Y} \end{matrix}$	115–125	125–135	135–145	145–155	155–165	165–175	Total
60–62	2	1	0	0	0	0	3
62–64	1	0	3	0	0	0	4
64–66	1	1	2	1	0	0	5
66–68	0	2	0	2	0	0	4
68–70	0	1	1	0	1	1	4
Total	4	5	6	3	1	1	20

D.

$\begin{matrix} \text{X} \\ \text{Y} \end{matrix}$	35–45	45–55	55–65	65–75	Total
115–130	4	0	0	0	4
130–145	2	1	3	0	6
145–160	1	1	2	3	7
160–175	1	2	2	2	7
Total	8	4	7	5	24

Correct answer:

$\begin{matrix} \text{X} \\ \text{Y} \end{matrix}$	0–4	5–9	10–14	15–19	Total
5–8	2	0	3	1	6
9–12	2	9	1	0	12
13–16	1	6	4	1	12

Total	5	15	8	2	30
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Explanation and steps

Classification gives the displayed table.

Question 2 - MCQ · PG-53 · MEX-Q1(i)

Following data gives the coded price (x) and demand (y) of a commodity.

Price	5	7	9	8	10	7	9	8	5	11	11	10	2	3	9
Demand	9	15	13	15	14	10	11	14	10	14	6	14	15	11	12
Price	2	4	3	14	6	10	7	15	8	6	5	6	11	14	15
Demand	6	11	8	11	10	15	9	15	13	9	14	10	7	5	6

You may use the bivariate frequency table given below.

X \ Y	0-4	5-9	10-14	15-19	Total
5-8	2	0	3	1	6
9-12	2	9	1	0	12
13-16	1	6	4	1	12
Total	5	15	8	2	30

Find marginal frequency distribution of x.

- A. 5, 15, 8, 2
- B. 6, 12, 12
- C. 7, 5, 9, 9
- D. 30, 30, 30, 30

Correct answer: 5, 15, 8, 2

Explanation and steps

Read the column totals.

Question 3 - MCQ · PG-53 · MEX-Q1(i)

Following data gives the coded price (x) and demand (y) of a commodity.

Price	5	7	9	8	10	7	9	8	5	11	11	10	2	3	9
Demand	9	15	13	15	14	10	11	14	10	14	6	14	15	11	12
Price	2	4	3	14	6	10	7	15	8	6	5	6	11	14	15
Demand	6	11	8	11	10	15	9	15	13	9	14	10	7	5	6

You may use the bivariate frequency table given below.

Y \ X	0–4	5–9	10–14	15–19	Total
5–8	2	0	3	1	6
9–12	2	9	1	0	12
13–16	1	6	4	1	12
Total	5	15	8	2	30

Find marginal frequency distribution of y.

- A. 6, 12, 12
- B. 5, 15, 8, 2
- C. 6, 10, 14
- D. 30, 30, 30

Correct answer: 6, 12, 12

Explanation and steps

Read the row totals.

Question 4 - MCQ · PG-53 · MEX-Q1(ii)

Following data gives the coded price (x) and demand (y) of a commodity.

Price	5	7	9	8	10	7	9	8	5	11	11	10	2	3	9
Demand	9	15	13	15	14	10	11	14	10	14	6	14	15	11	12
Price	2	4	3	14	6	10	7	15	8	6	5	6	11	14	15
Demand	6	11	8	11	10	15	9	15	13	9	14	10	7	5	6

You may use the bivariate frequency table given below.

Y \ X	0–4	5–9	10–14	15–19	Total
5–8	2	0	3	1	6
9–12	2	9	1	0	12
13–16	1	6	4	1	12
Total	5	15	8	2	30

Find conditional frequency distribution of y when x is less than 10.

- A. 2, 11, 7
- B. 6, 12, 12
- C. 5, 15, 8
- D. 2, 3, 1

Correct answer: 2, 11, 7

Explanation and steps

Add the first two X columns.

Question 5 - MCQ · PG-54 · MEX-Q2

Following data gives the age in years and marks obtained by 30 students in an intelligence test.

Age	16	17	22	19	21	16
Marks	16	19	39	50	48	41
Age	21	20	20	23	22	19
Marks	59	44	42	62	37	67
Age	23	20	22	22	23	22
Marks	45	57	35	37	38	56
Age	17	18	16	21	19	20
Marks	54	61	47	67	49	56
Age	17	18	23	21	20	16
Marks	51	42	65	56	52	48

Prepare a bivariate frequency distribution by taking class intervals 16–18, 18–20,...etc. for age and 10–20, 20–30... etc. for marks.

A.

Y \ X	16–18	18–20	20–22	22–24	Total
10–20	2	0	0	0	2
20–30	0	0	0	0	0
30–40	0	0	0	5	5
40–50	3	2	3	1	9
50–60	2	1	5	1	9
60–70	0	2	1	2	5
Total	7	5	9	9	30

B.

Y \ X	0–4	5–9	10–14	15–19	Total
5–8	2	0	3	1	6
9–12	2	9	1	0	12
13–16	1	6	4	1	12
Total	5	15	8	2	30

C.

Y \ X	115–125	125–135	135–145	145–155	155–165	165–175	Total
60–62	2	1	0	0	0	0	3
62–64	1	0	3	0	0	0	4
64–66	1	1	2	1	0	0	5
66–68	0	2	0	2	0	0	4
68–70	0	1	1	0	1	1	4
Total	4	5	6	3	1	1	20

D.

Y \ X	35–45	45–55	55–65	65–75	Total
115–130	4	0	0	0	4
130–145	2	1	3	0	6
145–160	1	1	2	3	7
160–175	1	2	2	2	7
Total	8	4	7	5	24

Correct answer:

X	16–18	18–20	20–22	22–24	Total
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Y					
10–20	2	0	0	0	2
20–30	0	0	0	0	0
30–40	0	0	0	5	5
40–50	3	2	3	1	9
50–60	2	1	5	1	9
60–70	0	2	1	2	5
Total	7	5	9	9	30

Explanation and steps

Classification gives the displayed table.

Question 6 - MCQ · PG-54 · MEX-Q2(i)

Following data gives the age in years and marks obtained by 30 students in an intelligence test.

Age	16	17	22	19	21	16
Marks	16	19	39	50	48	41
Age	21	20	20	23	22	19
Marks	59	44	42	62	37	67
Age	23	20	22	22	23	22
Marks	45	57	35	37	38	56
Age	17	18	16	21	19	20
Marks	54	61	47	67	49	56
Age	17	18	23	21	20	16
Marks	51	42	65	56	52	48

You may use the bivariate frequency table given below.

X Y	16–18	18–20	20–22	22–24	Total
10–20	2	0	0	0	2
20–30	0	0	0	0	0
30–40	0	0	0	5	5
40–50	3	2	3	1	9
50–60	2	1	5	1	9
60–70	0	2	1	2	5
Total	7	5	9	9	30

Find marginal frequency distribution of age.

- A. 7, 5, 9, 9
- B. 2, 0, 5, 9, 9, 5
- C. 5, 15, 8, 2
- D. 30, 30, 30, 30

Correct answer: 7, 5, 9, 9

Explanation and steps

Read the column totals.

Question 7 - MCQ · PG-54 · MEX-Q2(i)

Following data gives the age in years and marks obtained by 30 students in an intelligence test.

Age	16	17	22	19	21	16
Marks	16	19	39	50	48	41
Age	21	20	20	23	22	19
Marks	59	44	42	62	37	67
Age	23	20	22	22	23	22
Marks	45	57	35	37	38	56
Age	17	18	16	21	19	20
Marks	54	61	47	67	49	56
Age	17	18	23	21	20	16
Marks	51	42	65	56	52	48

You may use the bivariate frequency table given below.

X \ Y	16–18	18–20	20–22	22–24	Total
10–20	2	0	0	0	2
20–30	0	0	0	0	0
30–40	0	0	0	5	5
40–50	3	2	3	1	9
50–60	2	1	5	1	9
60–70	0	2	1	2	5
Total	7	5	9	9	30

Find marginal frequency distribution of marks.

- A. 2, 0, 5, 9, 9, 5
- B. 7, 5, 9, 9
- C. 2, 5, 9, 9, 5, 0
- D. 30, 30, 30, 30, 30, 30

Correct answer: 2, 0, 5, 9, 9, 5

Explanation and steps

Read the row totals.

Question 8 - MCQ · PG-54 · MEX-Q2(ii)

Following data gives the age in years and marks obtained by 30 students in an intelligence test.

Age	16	17	22	19	21	16
Marks	16	19	39	50	48	41
Age	21	20	20	23	22	19
Marks	59	44	42	62	37	67
Age	23	20	22	22	23	22
Marks	45	57	35	37	38	56
Age	17	18	16	21	19	20
Marks	54	61	47	67	49	56
Age	17	18	23	21	20	16
Marks	51	42	65	56	52	48

You may use the bivariate frequency table given below.

X \ Y	16–18	18–20	20–22	22–24	Total
10–20	2	0	0	0	2
20–30	0	0	0	0	0
30–40	0	0	0	5	5
40–50	3	2	3	1	9
50–60	2	1	5	1	9
60–70	0	2	1	2	5
Total	7	5	9	9	30

Find conditional frequency distribution of marks obtained when age of students is between 20–22.

- A. 0, 0, 0, 3, 5, 1
- B. 2, 0, 5, 9, 9, 5
- C. 0, 0, 3, 5, 1, 0
- D. 7, 5, 9, 9

Correct answer: 0, 0, 0, 3, 5, 1

Explanation and steps

Read the 20–22 age column.

Question 9 - MCQ · PG-54 · MEX-Q3(i)

Following data gives Sales (in Lakh Rs.) and Advertisement Expenditure (in Thousand Rs.) of 20 firms.

(115, 61), (120, 60), (128, 61), (121, 63), (137, 62), (139, 62), (143, 63), (117, 65), (126, 64), (141, 65), (140, 65), (153, 64), (129, 67), (130, 66), (150, 67), (148, 66), (130, 69), (138, 68), (155, 69), (172, 68)

Construct a bivariate frequency distribution table for the above data by taking classes 115–125, 125–135,etc. for sales and 60–62, 62–64, ...etc. for advertisement expenditure.

A.

$\begin{matrix} X \\ Y \end{matrix}$	115–125	125–135	135–145	145–155	155–165	165–175	Total
60–62	2	1	0	0	0	0	3
62–64	1	0	3	0	0	0	4
64–66	1	1	2	1	0	0	5
66–68	0	2	0	2	0	0	4
68–70	0	1	1	0	1	1	4
Total	4	5	6	3	1	1	20

B.

$\begin{matrix} X \\ Y \end{matrix}$	16–18	18–20	20–22	22–24	Total
10–20	2	0	0	0	2
20–30	0	0	0	0	0
30–40	0	0	0	5	5
40–50	3	2	3	1	9
50–60	2	1	5	1	9
60–70	0	2	1	2	5
Total	7	5	9	9	30

C.

$\begin{matrix} X \\ Y \end{matrix}$	35–45	45–55	55–65	65–75	Total
115–130	4	0	0	0	4
130–145	2	1	3	0	6
145–160	1	1	2	3	7
160–175	1	2	2	2	7
Total	8	4	7	5	24

D.

$\begin{matrix} X \\ Y \end{matrix}$	80–90	90–100	100–110	110–120	120–130	Total
500–600	0	1	0	2	1	4
600–700	2	2	2	1	0	7
700–800	1	4	0	0	1	6
800–900	1	3	1	0	3	8
900–1000	3	1	0	1	0	5
Total	7	11	3	4	5	30

Correct answer:

$\begin{matrix} X \\ Y \end{matrix}$	115–125	125–135	135–145	145–155	155–165	165–175	Total
60–62	2	1	0	0	0	0	3
62–64	1	0	3	0	0	0	4
64–66	1	1	2	1	0	0	5
66–68	0	2	0	2	0	0	4
68–70	0	1	1	0	1	1	4

Total	4	5	6	3	1	1	20
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Explanation and steps

Classification gives the displayed table.

Question 10 - MCQ · PG-54 · MEX-Q3(ii)

Following data gives Sales (in Lakh Rs.) and Advertisement Expenditure (in Thousand Rs.) of 20 firms.

(115, 61), (120, 60), (128, 61), (121, 63), (137, 62), (139, 62), (143, 63), (117, 65), (126, 64), (141, 65), (140, 65), (153, 64), (129, 67), (130, 66), (150, 67), (148, 66), (130, 69), (138, 68), (155, 69), (172, 68)

You may use the bivariate frequency table given below.

X \ Y	115–125	125–135	135–145	145–155	155–165	165–175	Total
60–62	2	1	0	0	0	0	3
62–64	1	0	3	0	0	0	4
64–66	1	1	2	1	0	0	5
66–68	0	2	0	2	0	0	4
68–70	0	1	1	0	1	1	4
Total	4	5	6	3	1	1	20

Find marginal frequency distribution of sales.

- A. 4, 5, 6, 3, 1, 1
- B. 3, 4, 5, 4, 4
- C. 4, 6, 5, 3, 1, 1
- D. 20, 20, 20, 20, 20, 20

Correct answer: 4, 5, 6, 3, 1, 1

Explanation and steps

Read the column totals.

Question 11 - MCQ · PG-54 · MEX-Q3(ii)

Following data gives Sales (in Lakh Rs.) and Advertisement Expenditure (in Thousand Rs.) of 20 firms.

(115, 61), (120, 60), (128, 61), (121, 63), (137, 62), (139, 62), (143, 63), (117, 65), (126, 64), (141, 65), (140, 65), (153, 64), (129, 67), (130, 66), (150, 67), (148, 66), (130, 69), (138, 68), (155, 69), (172, 68)

You may use the bivariate frequency table given below.

Y \ X	115–125	125–135	135–145	145–155	155–165	165–175	Total
60–62	2	1	0	0	0	0	3
62–64	1	0	3	0	0	0	4
64–66	1	1	2	1	0	0	5
66–68	0	2	0	2	0	0	4
68–70	0	1	1	0	1	1	4
Total	4	5	6	3	1	1	20

Find marginal frequency distribution of advertisement expenditure.

- A. 3, 4, 5, 4, 4
- B. 4, 5, 6, 3, 1, 1
- C. 3, 5, 4, 4, 4
- D. 20, 20, 20, 20, 20

Correct answer: 3, 4, 5, 4, 4

Explanation and steps

Read the row totals.

Question 12 - MCQ · PG-54 · MEX-Q3(iii)

Following data gives Sales (in Lakh Rs.) and Advertisement Expenditure (in Thousand Rs.) of 20 firms.

(115, 61), (120, 60), (128, 61), (121, 63), (137, 62), (139, 62), (143, 63), (117, 65), (126, 64), (141, 65), (140, 65), (153, 64), (129, 67), (130, 66), (150, 67), (148, 66), (130, 69), (138, 68), (155, 69), (172, 68)

You may use the bivariate frequency table given below.

Y \ X	115–125	125–135	135–145	145–155	155–165	165–175	Total
60–62	2	1	0	0	0	0	3
62–64	1	0	3	0	0	0	4
64–66	1	1	2	1	0	0	5
66–68	0	2	0	2	0	0	4
68–70	0	1	1	0	1	1	4
Total	4	5	6	3	1	1	20

Find conditional frequency distribution of Sales when the advertisement expenditure is between 64–66 (Thousand Rs.).

- A. 1, 1, 2, 1, 0, 0
- B. 4, 5, 6, 3, 1, 1
- C. 1, 0, 1, 2, 1, 0
- D. 3, 4, 5, 4, 4

Correct answer: 1, 1, 2, 1, 0, 0

Explanation and steps

Read the 64–66 row.

Question 13 - MCQ · PG-54 · MEX-Q3(iv)

Following data gives Sales (in Lakh Rs.) and Advertisement Expenditure (in Thousand Rs.) of 20 firms.

(115, 61), (120, 60), (128, 61), (121, 63), (137, 62), (139, 62), (143, 63), (117, 65), (126, 64), (141, 65), (140, 65), (153, 64), (129, 67), (130, 66), (150, 67), (148, 66), (130, 69), (138, 68), (155, 69), (172, 68)

You may use the bivariate frequency table given below.

Y \ X	115–125	125–135	135–145	145–155	155–165	165–175	Total
60–62	2	1	0	0	0	0	3
62–64	1	0	3	0	0	0	4
64–66	1	1	2	1	0	0	5
66–68	0	2	0	2	0	0	4
68–70	0	1	1	0	1	1	4
Total	4	5	6	3	1	1	20

Find conditional frequency distribution of advertisement expenditure when the sales are between 125–135 (Lakh Rs.).

- A. 1, 0, 1, 2, 1
- B. 3, 4, 5, 4, 4
- C. 1, 1, 2, 1, 0
- D. 4, 5, 6, 3, 1

Correct answer: 1, 0, 1, 2, 1

Explanation and steps

Read the 125–135 column.

Question 14 - MCQ · PG-54 · MEX-Q4

Prepare a bivariate frequency distribution for the following data, taking class intervals for X as 35–45, 45–55...etc and for Y as 115–130, 130–145...etc. where, X denotes the age in years and Y denotes blood pressure for a group of 24 persons.

(55, 151), (36, 140), (72, 160), (38, 124), (65, 148), (46, 130), (58, 152), (50, 149), (38, 115), (42, 145), (41, 163), (47, 161), (69, 159), (60, 161), (58, 131), (57, 136), (43, 141), (52, 164), (59, 161), (44, 128), (35, 118), (62, 142), (67, 157), (70, 162)

A.

Y \ X	35–45	45–55	55–65	65–75	Total
115–130	4	0	0	0	4
130–145	2	1	3	0	6
145–160	1	1	2	3	7
160–175	1	2	2	2	7
Total	8	4	7	5	24

B.

Y \ X	115–125	125–135	135–145	145–155	155–165	165–175	Total
60–62	2	1	0	0	0	0	3
62–64	1	0	3	0	0	0	4
64–66	1	1	2	1	0	0	5
66–68	0	2	0	2	0	0	4
68–70	0	1	1	0	1	1	4
Total	4	5	6	3	1	1	20

C.

Y \ X	16–18	18–20	20–22	22–24	Total
10–20	2	0	0	0	2
20–30	0	0	0	0	0
30–40	0	0	0	5	5
40–50	3	2	3	1	9
50–60	2	1	5	1	9
60–70	0	2	1	2	5
Total	7	5	9	9	30

D.

Y \ X	80–90	90–100	100–110	110–120	120–130	Total
500–600	0	1	0	2	1	4
600–700	2	2	2	1	0	7
700–800	1	4	0	0	1	6
800–900	1	3	1	0	3	8
900–1000	3	1	0	1	0	5
Total	7	11	3	4	5	30

Correct answer:

Y \ X	35–45	45–55	55–65	65–75	Total
115–130	4	0	0	0	4
130–145	2	1	3	0	6
145–160	1	1	2	3	7
160–175	1	2	2	2	7

Total	8	4	7	5	24
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Explanation and steps

Classification gives the displayed table.

Question 15 - MCQ · PG-54 · MEX-Q4(i)

Prepare a bivariate frequency distribution for the following data, taking class intervals for X as 35–45, 45–55...etc and for Y as 115–130, 130–145...etc. where, X denotes the age in years and Y denotes blood pressure for a group of 24 persons.

(55, 151), (36, 140), (72, 160), (38, 124), (65, 148), (46, 130), (58, 152), (50, 149), (38, 115), (42, 145), (41, 163), (47, 161), (69, 159), (60, 161), (58, 131), (57, 136), (43, 141), (52, 164), (59, 161), (44, 128), (35, 118), (62, 142), (67, 157), (70, 162)

You may use the bivariate frequency table given below.

X \ Y	35–45	45–55	55–65	65–75	Total
115–130	4	0	0	0	4
130–145	2	1	3	0	6
145–160	1	1	2	3	7
160–175	1	2	2	2	7
Total	8	4	7	5	24

Find marginal frequency distribution of X.

- A. 8, 4, 7, 5
- B. 4, 6, 7, 7
- C. 8, 7, 4, 5
- D. 24, 24, 24, 24

Correct answer: 8, 4, 7, 5

Explanation and steps

Read the column totals.

Question 16 - MCQ · PG-54 · MEX-Q4(ii)

Prepare a bivariate frequency distribution for the following data, taking class intervals for X as 35–45, 45–55...etc and for Y as 115–130, 130–145...etc. where, X denotes the age in years and Y denotes blood pressure for a group of 24 persons.

(55, 151), (36, 140), (72, 160), (38, 124), (65, 148), (46, 130), (58, 152), (50, 149), (38, 115), (42, 145), (41, 163), (47, 161), (69, 159), (60, 161), (58, 131), (57, 136), (43, 141), (52, 164), (59, 161), (44, 128), (35, 118), (62, 142), (67, 157), (70, 162)

You may use the bivariate frequency table given below.

Y \ X	35–45	45–55	55–65	65–75	Total
115–130	4	0	0	0	4
130–145	2	1	3	0	6
145–160	1	1	2	3	7
160–175	1	2	2	2	7
Total	8	4	7	5	24

Find conditional frequency distribution of Y when $X < 45$.

- A. 4, 2, 1, 1
- B. 4, 6, 7, 7
- C. 8, 4, 7, 5
- D. 0, 2, 1, 4

Correct answer: 4, 2, 1, 1

Explanation and steps

Read the 35–45 column.

Question 17 - MCQ · PG-54 · MEX-Q5

Thirty pairs of values of two variables X and Y are given below.

X	110	88	91	115	97	85	85	91	120	95
Y	500	800	870	599	625	650	905	700	850	824
X	82	105	99	90	108	124	90	90	111	89
Y	970	609	990	735	600	735	729	840	999	780
X	112	100	87	92	91	82	96	120	121	122
Y	638	850	630	720	695	923	555	810	805	526

Form a bivariate frequency table.

A.

Y \ X	80–90	90–100	100–110	110–120	120–130	Total
500–600	0	1	0	2	1	4
600–700	2	2	2	1	0	7
700–800	1	4	0	0	1	6
800–900	1	3	1	0	3	8
900–1000	3	1	0	1	0	5
Total	7	11	3	4	5	30

B.

Y \ X	35–45	45–55	55–65	65–75	Total
115–130	4	0	0	0	4
130–145	2	1	3	0	6
145–160	1	1	2	3	7
160–175	1	2	2	2	7
Total	8	4	7	5	24

C.

Y \ X	115–125	125–135	135–145	145–155	155–165	165–175	Total
60–62	2	1	0	0	0	0	3
62–64	1	0	3	0	0	0	4
64–66	1	1	2	1	0	0	5
66–68	0	2	0	2	0	0	4
68–70	0	1	1	0	1	1	4
Total	4	5	6	3	1	1	20

D.

Y \ X	0–4	5–9	10–14	15–19	Total
5–8	2	0	3	1	6
9–12	2	9	1	0	12
13–16	1	6	4	1	12
Total	5	15	8	2	30

Correct answer:

Y \ X	80–90	90–100	100–110	110–120	120–130	Total
500–600	0	1	0	2	1	4
600–700	2	2	2	1	0	7
700–800	1	4	0	0	1	6
800–900	1	3	1	0	3	8

900–1000	3	1	0	1	0	5
Total	7	11	3	4	5	30

Explanation and steps

Classification gives the displayed table.

Question 18 - MCQ · PG-54 · MEX-Q5

Thirty pairs of values of two variables X and Y are given below.

X	110	88	91	115	97	85	85	91	120	95
Y	500	800	870	599	625	650	905	700	850	824
X	82	105	99	90	108	124	90	90	111	89
Y	970	609	990	735	600	735	729	840	999	780
X	112	100	87	92	91	82	96	120	121	122
Y	638	850	630	720	695	923	555	810	805	526

You may use the bivariate frequency table given below.

X \ Y	80–90	90–100	100–110	110–120	120–130	Total
500–600	0	1	0	2	1	4
600–700	2	2	2	1	0	7
700–800	1	4	0	0	1	6
800–900	1	3	1	0	3	8
900–1000	3	1	0	1	0	5
Total	7	11	3	4	5	30

Find marginal frequency distribution of X.

- A. 7, 11, 3, 4, 5
- B. 4, 7, 6, 8, 5
- C. 7, 3, 11, 4, 5
- D. 30, 30, 30, 30, 30

Correct answer: 7, 11, 3, 4, 5

Explanation and steps

Read the column totals.

Question 19 - MCQ · PG-54 · MEX-Q5

Thirty pairs of values of two variables X and Y are given below.

X	110	88	91	115	97	85	85	91	120	95
Y	500	800	870	599	625	650	905	700	850	824
X	82	105	99	90	108	124	90	90	111	89
Y	970	609	990	735	600	735	729	840	999	780
X	112	100	87	92	91	82	96	120	121	122
Y	638	850	630	720	695	923	555	810	805	526

You may use the bivariate frequency table given below.

X \ Y	80–90	90–100	100–110	110–120	120–130	Total
500–600	0	1	0	2	1	4
600–700	2	2	2	1	0	7
700–800	1	4	0	0	1	6
800–900	1	3	1	0	3	8
900–1000	3	1	0	1	0	5
Total	7	11	3	4	5	30

Find marginal frequency distribution of Y.

- A. 4, 7, 6, 8, 5
- B. 7, 11, 3, 4, 5
- C. 4, 6, 7, 8, 5
- D. 30, 30, 30, 30, 30

Correct answer: 4, 7, 6, 8, 5

Explanation and steps

Read the row totals.

Question 20 - ENTER ANSWER · PG-54 · MEX-Q6

Following table shows how the samples of Mathematics and Economics scores of 25 students are distributed:

Marks in Economics \ Marks in Mathematics	40–70	70–100
40–70	20	15
70–100	5	10

Find the value of χ^2 statistic.

Correct answer: 2.38

Explanation and steps

Using the observed table and expected frequencies gives $\chi^2 \approx 2.38$.

Question 21 - ENTER ANSWER · PG-54 · MEX-Q7

Compute χ^2 statistic from following data:

	Graduates	Post-Graduates
Male	28	22
Female	32	18

Correct answer: 0.67

Explanation and steps

The four cell contributions total approximately 0.67.

Question 22 - ENTER ANSWER · PG-54–55 · MEX-Q8

Attitude of 250 employees towards a proposed policy of the company is observed in the following table.

	Favor	Indifferent	Oppose
Male	68	46	36
Female	27	49	24

Calculate χ^2 statistic.

Correct answer: 10.614

Explanation and steps

The six cell contributions total approximately 10.614.

Question 23 - ENTER ANSWER · PG-55 · MEX-Q9

In a certain sample of 1000 families, 450 families are consumers of tea. Out of 600 Hindu families, 286 families consume tea.

Calculate χ^2 statistic.

Correct answer: 4.31

Explanation and steps

Complete the 2×2 table and sum the four chi-square contributions.

Question 24 - ENTER ANSWER · PG-55 · MEX-Q10

A sample of boys and girls were asked to choose their favourite sport, with the following results.

	Football	Cricket	Hockey	Basketball
Boys	86	60	44	10
Girls	40	30	25	5

Find the value of χ^2 statistic.

Correct answer: 0.4076

Explanation and steps

The eight cell contributions give $\chi^2 \approx 0.4076$.